

Assignment 3 – Architecture Diagram Explanation

AI Engineering Fellowship 2026 – Week 3

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1. User

The user interacts with the Personal Productivity Agent through the frontend. Users can create, update, complete, delete, and view tasks, save and search notes, extract meeting actions, and generate work plans.

2. Frontend

The frontend provides the user interface where requests are entered and results are displayed. It sends user requests to the Agent API and presents responses, approval prompts, and errors.

3. Agent API

The Agent API receives requests from the frontend, initializes the agent workflow, and coordinates communication between the interface, agent state, language model, and tools.

4. Agent State

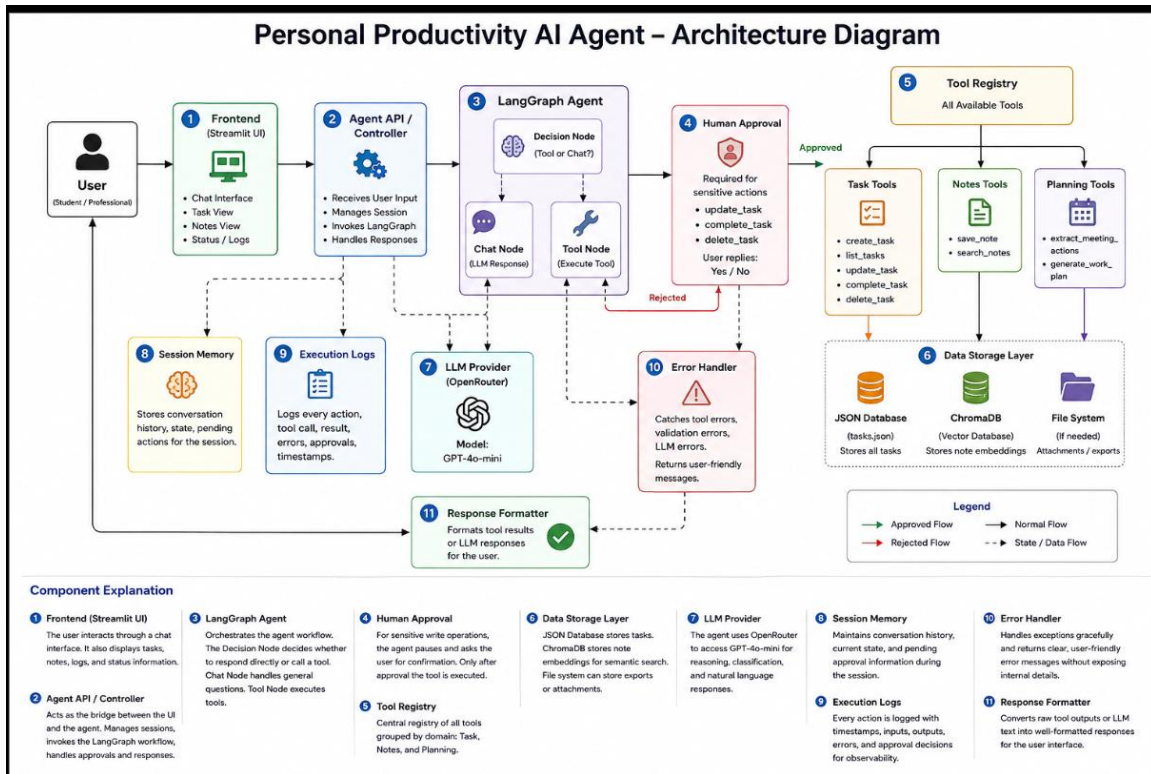
The Agent State stores conversation history, pending approval actions, tool outputs, execution status, and other session information required during a conversation.

5. Model Provider (LLM)

The Large Language Model analyzes user requests, decides whether tools are required, selects the appropriate tool, and generates conversational responses when no tool is needed.

6. Tool Registry

The Tool Registry maintains all available tools and dispatches execution requests to the correct tool based on the agent's decision.



7. Task Tools

These tools manage productivity tasks, including creating, listing, updating, completing, and deleting tasks.

8. Notes Tools

These tools save notes and perform semantic note search using the vector database.

9. Planning Tools

Planning tools extract meeting action items and generate daily work plans from tasks and priorities.

10. Database

The database stores tasks, notes, metadata, and other persistent information used by the agent.

11. Human Approval

Sensitive write operations such as updating, completing, and deleting tasks require explicit user approval before execution.

12. Error Handling

Errors such as invalid tool arguments, missing tasks, or execution failures are caught and converted into clear user-friendly responses.

13. Session Memory

The application preserves conversation context during the current session, allowing the agent to continue multi-step interactions.

14. Execution Logs / Observability

Execution logs record user requests, selected tools, approvals, tool execution results, and errors to support debugging and evaluation.